X. Properties of a Random Sample (GSE 518)

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We introduce the goal and some preliminary concepts and ideas from statistics.

1. Statistics: Introduction

The goal of statistics and econometrics is to draw inferences about a population based on a sample of the population.

- Say we take a sample $(X_1, X_2, \ldots, X_n)$ from $X \sim f_X(x)$, where $f_X(x)$ is the distribution of X in the *population*.
  - Example: Get data on wages of $1,000$ US workers: $(X_1, X_2, \ldots, X_{1,000})$

$$\begin{array}{c} X = \text{wages and } X \sim f_X(x), \\ \text{the distribution of wages in the} \\ \text{population (e.g., all U.S. workers)} \end{array} \longrightarrow \begin{array}{c} (X_1, X_2, \ldots, X_{1000}) \\ \text{Our sample} \end{array}$$

  - **Key question:** What can you learn about $f_X(x)$ in a population based on our sample $(X_1, \ldots, X_n)$?
    - Can we learn, e.g., about $E(X)$, $Var(X)$, $Pr(X \leq r)$, entire $f_X(x)$?
  - **Goal:** To use our sample to learn about $f_X(x)$ or features of $f_X(x)$.

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There are two general approaches:

1. Non-parametric Approach:

- Let $f_X(x)$ be any unknown distribution.
- What can we learn? $E(X) =?$ $\text{Var}(X) =?$ $f_X(x)?$

2. Parametric Approach:

- Here, we assume X comes from a specific $f_X(x)$, where the distribution is *known* but its parameters are *unknown*.
- For example: $X \sim N(\mu, \sigma^2)$, where μ and σ^2 are unknown. Or, we assume $X \sim Gamma(\alpha, \beta)$, where α and β are unknown.
- Here, our goal is to estimate (or "guess") the values of the parameters (e.g.μ and σ^2) using our sample $(X_1, X_2, \ldots, X_n)$.
- Note that once we have our estimated parameters, then we "have" the entire distribution $f_X(x)$ and hence we can get any related object. For example, any moment (e.g., $E(X)$, $Var(X)$) or probability (e.g., $Pr(X \leq r)$.

Note: The more we assume, the more we get.

- *Cost:* If our assumptions are wrong, our conclusions are wrong.

Next:

- Basic Statistics Concepts
- Look at two common data summaries: sample mean ($\overline{X}$) and sample variance (S^2).

Note: Here we do everything with a single random variable X. Later in econometrics you will apply these ideas to a multivariate case (e.g., work with $f_{Y|X}(y|x)$, $E(Y|X)$, etc).

2. Some Key Statistics Concepts

Definition: *(Def. 5.1.1. in textbook.)* $(X_1, \ldots, X_n)$ are a *random sample* of size n from the distribution $f_X(x)$ (population) if:

1. $(X_1, X_2, \ldots, X_n)$ are mutually independent. *(Independent)*
2. Each X_i comes from the same distribution, i.e., $X_i \sim f_X(x)$. *(Identically Distributed)*

- Notes on "Random Sample":

- Also called an *i.i.d. sample.*
- n is the sample size.
- The joint density of $(X_1, X_2, \ldots, X_n)$ is:

$$f_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdots f_{X_n}(x_n) \quad \text{(product of marginals, by independence)}$$

$$= \prod_{i=1}^{n} f_X(x_i) \quad \text{(since they are identically distributed)}$$

Definition: *(Def. 5.2.1. in textbook.)* Given a sample $(X_1, \ldots, X_n)$, a *statistic* is any real-valued function of $(X_1, \ldots, X_n)$, say, $Y = T(X_1, \ldots, X_n)$. The distribution of the statistic is called the *sampling distribution* of Y.

Notes on "Statistic":

- Examples:

1. Sample mean:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} x_i = T(X_1, \ldots, X_n)$$

2. Sample variance:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

3. Sample standard deviation:

$$S = \sqrt{S^2}$$

4. Sample covariance:

$$S_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$$

5. Sample correlation:

$$\hat{\rho}_{XY} = \frac{S_{XY}}{S_X S_Y}$$

- $(X_1, \ldots, X_n)$ are random variables, hence the statistic $Y = T(X_1, \ldots, X_n)$ *is also a random variable.*

  - Hence, Y (i.e., any statistic) has a distribution, $Y \sim f_Y(y)$, and consequently, $E(Y)$, $Var(Y)$, cdf $F_Y(y)$, etc.
  - Intuitively, why is Y random (e.g., the sample mean $\overline{X}$)? Because its value depends on the *specific* sample we draw.

- Key Note: The (population or true) mean $E(X)$ is fixed (i.e., non-random) but is unknown. The sample mean $\overline{x}$ is random).

3. Analyzing $\overline{X}$ and S^2

Here, we look at two common statistics: sample mean ($\overline{X}$) and sample variance (S^2), where

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

Notation: We use *capital letters* to denote random variables—e.g., $(X_1, X_2, \ldots, X_n)$—and *lowercase letters* to denote specific values–e.g., $(x_1, x_2, \ldots, x_n)$.

3.1. Case 1: Non-parametric (for any known or unknown $f_X(x)$)

Useful facts about $\overline{X}$ and S^2 : *(Thm. 5.2.4. in textbook.)*

a). $\overline{X}$ minimizes the sum of squared errors:

$$\overline{X} = \arg\min_a \sum_{i=1}^{n} (X_i - a)^2$$

- *Proof of part (a):* See lecture.
- *Intuition:* Say we draw another value of X and we want to guess its value based on our sample $(X_1, X_2, \ldots, X_n)$. We ask: Find the *best* guess of its value (call it a) in the sense of minimizing the sum of square errors.
  * Our best guess a is the sample mean $\overline{X}$.

b). $(n-1)S^2 = \sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2$

- *Proof of part (b):*

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n}\left[(X_i^2 - 2X_i\overline{X} + \overline{X}^2\right] = \sum_{i=1}^{n} X_i^2 - 2\overline{X}\sum_{i=1}^{n} x_i + n\overline{X}^2$$

Note that: $\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n} \Rightarrow \sum_{i=1}^{n} X_i = n\overline{X}$. Hence, $2\overline{X}\sum_{i=1}^{n} x_i = 2n\overline{X}^2$. Plugging this in the equation above, we find:

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - 2n\overline{X}^2 + n\overline{X}^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2$$

Theorem 3.1: *(Thm. 5.2.6. in textbook.)* Let $X_1, \ldots, X_n$ be an iid sample from a population with mean $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Then:

a). $E(\overline{X}) = \mu$

b). $Var(\overline{X}) = \frac{\sigma^2}{n}$

c). $E(S^2) = \sigma^2$

- *Proof of part (a):* To show: $E(\overline{X}) = \mu$

$$E(\overline{X}) = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}E\left[\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\cdot n\mu = \mu$$

  - Hence, $f_{\overline{X}}(x)$ is centered at $E(X)$. In words, regardless of the distribution the data come from, the distribution of the sample mean ($\overline{X}$) is centered at $E(X)$.

- *Proof of part (b):* To show: $Var(\overline{X}) = \frac{\sigma^2}{n}$

$$Var(\overline{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}Var\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n^2}\cdot n(\sigma^2) = \frac{\sigma^2}{n}$$

  Since our sample is independent and identically distributed, $Cov(X_i, X_j) = 0$ for all $i \neq j$.

  - In words, regardless of the distribution the data come from, the variance (a measure of dispersion of the distribution of the sample mean, $\overline{X}$) equals the variance of the distribution X over the sample size.

- *Proof of part (c):* To show: $E(S^2) = \sigma^2$

$$E(S^2) = E\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = \frac{1}{n-1}E\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = \frac{1}{n-1}E\left[\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right]$$

The last equality in the line above uses part (b) of the "Useful facts about $\overline{X}$ and S^2" at the start of section 3.1. Distributing the expected value, we can write:

$$E(S^2) = \frac{1}{n-1}\sum_{i=1}^{n} E\left[X_i^2\right] - nE\left[\overline{X}^2\right] \tag{1}$$

While we do not directly have these two expectations, the usual variance formula $Var(Y) = E(Y^2) - (E(Y))^2$ implies that $E(Y^2) = Var(Y) + (E(Y))^2$. Using this equation, along with results (a) and (b) from Theorem 3.1., we find:

$$E(X_i^2) = Var(X_i) + (E(X_i))^2 = \sigma^2 + \mu^2$$

$$E(\overline{X}^2) = Var(\overline{X}) + (E(\overline{X}))^2 = \frac{\sigma^2}{n} + \mu^2$$

Finally, plugging these two expectations into equation (1), we find

$$\begin{aligned} E(S^2) &= \frac{1}{n-1}\left[\sum_{i=1}^{n}(\sigma^2+\mu^2) - n\left(\frac{\sigma^2}{n}+\mu^2\right)\right] \\ &= \frac{1}{n-1}\left[n(\sigma^2+\mu^2) - n\left(\frac{\sigma^2}{n}+\mu^2\right)\right] \\ &= \frac{1}{n-1}\left[n\sigma^2 + n\mu^2 - n\sigma^2 + \mu^2\right] = \frac{(n-1)\sigma^2}{(n-1)} \\ &= \sigma^2 \end{aligned}$$

- Hence, $f_{S^2}(s)$ is centered at $Var(X)$. In words, regardless of the distribution the data come from, the distribution of the sample variance ($\S^2$) is centered at $Var(X)$.

Notes on Theorem 3.1:

- Given that $E(\overline{X}) = \mu$ and $E(S^2) = \sigma^2$, we say that $\overline{X}$ and σ^2 are *unbiased* estimators of μ and σ^2, respectively (later).

- Recall the sample standard deviation $S = \sqrt{S^2}$.

  - Does $E(S^2) = \sigma^2$ imply $E(S) = \sigma$?

  - If not: (i) Why? (ii) Could we say if $E(S) > \sigma$ (overestimate) or $E(S) < \sigma$ (underestimate)?

    (See lecture for details.)

- Without assuming a specific pdf $f_X(x)$ for X it is not possible to derive the *exact* distribution of $\overline{X}$ and σ^2.

  - However, we can get an *approximate* distribution as $n \to \infty$ (later in "asymptotics").

3.2. Case 2: Parametric Case (assume $f_X(x)$ is known)

We now assume X comes from a specific distribution $f_X(x)$, where the distribution is *known* but its parameters are *unknown*.

- $E(\overline{X})$, $Var(\overline{X})$, $E(S^2)$ are the same as before in Section 3.1 (those results were for known or unknown $f_X(x)$).

- If $f_X(x)$ is assumed to be known, in principle, we can get the exact distribution of $\overline{X}$ and S^2 (using the mgf, or change of variable formula).

- A useful theorem to get the *exact* distribution of $\overline{X}$ in soem cases is the following:

Theorem 3.2: *(Thm. 5.2.7 from textbook.)* If $X_1, \ldots, X_n$ is an iid sample from $f_X(x)$ with mgf $M_X(t)$. Then,

$$M_{\overline{X}}(t) = \left(M_X\left(\frac{t}{n}\right)\right)^n$$

- *Proof:*

$$\begin{aligned} M_{\overline{X}}(t) = E\left(e^{t\overline{X}}\right) &= E\left(e^{t\left(\frac{\sum_{i=1}^n X_i}{n}\right)}\right) \quad \text{(by definition of mgf and } \overline{X}) \\ &= E\left(e^{\frac{t}{n}X_1} \cdot e^{\frac{t}{n}X_2} \cdots e^{\frac{t}{n}X_n}\right) \\ &= E\left(e^{\frac{t}{n}X_1}\right) \cdot E\left(e^{\frac{t}{n}X_2}\right) \cdots E\left(e^{\frac{t}{n}X_n}\right) \quad \text{(by } \textit{independence}) \\ &= \prod_{i=1}^n E\left(e^{\frac{t}{n}X_i}\right) = \prod_{i=1}^n M_{X_i}\left(t/n\right) \quad \text{(by definition of mgf)} \\ &= \left(M_X\left(t/n\right)\right)^n \quad \text{(by } \textit{identically distributed}\text{, all } X_i \text{ have same mgf)} \end{aligned}$$

In the next section, we focus on a common case: Assume $X \sim N(\mu, \sigma^2)$

3.2.1. Sampling from a Normal Distribution

- In this section 3.2.1., we let $X_1, \ldots, X_n$ be an iid sample from $N(\mu, \sigma^2)$.

  * For shorthand, we may write $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

- Under this assumption, the *goal* for the rest of these notes is to:

  1. Get the *exact* distribution of $\overline{X}$ and S^2.

  2. Show their relation to the t and F distributions.

Theorem 3.3: *(Thm. 5.3.1. in book.)* Let $X_1, \ldots, X_n$ be an iid sample from $N(\mu, \sigma^2)$. Then:

a $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$

b $\overline{X}$ and S^2 are independent

c $(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{(n-1)}$

- *Proof of part (a):* To show: $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

  Given $X_i \sim N(\mu, \sigma^2)$, we know the mgf of X_i is given by

$$M_{X_i}(t) = e^{\left[\mu t + \sigma^2 \cdot \frac{t^2}{2}\right]}$$

  Applying Theorem 3.2 above, we find

$$M_{\overline{X}}(t) = \left(M_{X_i}\left(\frac{t}{n}\right)\right)^n = \left[e^{\left[\mu\frac{t}{n} + \frac{\sigma^2}{2}\left(\frac{t}{n}\right)^2\right]}\right]^n$$

$$= e^{\left[\mu t + \frac{\sigma^2}{n}\cdot\frac{t^2}{2}\right]},$$

  which is the mgf of a $N\left(\mu, \frac{\sigma^2}{n}\right)$. Therefore, this implies that

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- *Proof of part (b):* To show: $\overline{X}$ and S^2 are independent.

  Skipped (a lot of algebra). The proof uses tools we have previously study: transformation of random variables and factorization theorem. (If interested, see textbook for details.)

– *Important Corollary to Theorem 3.3.(b):* $\overline{X}$ and the vector $(X_1-\overline{X}, X_2-\overline{X}, \ldots, X_n-\overline{X})$ are *independent*.

- *Proof of part (c):* To show:

$$(n-1)\frac{s^2}{\sigma^2} = \frac{(n-1)\sum_{i=1}^n (X_i-\overline{X})^2}{\sigma^2(n-1)} = \boxed{\frac{\sum_{i=1}^n (X_i-\overline{X})^2}{\sigma^2} \sim \chi^2_{(n-1)}}$$

Recall the following facts we have seen before:

1. If $Y_1, \ldots, Y_m$ are independent random variables with $Y_i \sim \chi^2_{(p_i)} \implies \sum_{i=1}^m Y_i \sim \chi^2_{(\sum_{i=1}^m p_i)}$ (see problem set 6 question 7(c))
2. If $Y_i \sim N(\mu, \sigma^2) \implies Y_i^2 \sim \chi^2_{(1)}$ (see problem set 5 question 5(b))
3. If $Y_1, \ldots, Y_n$ are independent and identically distributed random variables with $Y_i \sim N(0,1) \implies \sum_{i=1}^n Y_i^2 \sim \chi^2_{(n)}$ (combining the two results above)

In addition, we know that:

(i) $X_i \sim N(\mu, \sigma^2) \implies \frac{X_i-\mu}{\sigma} \sim N(0,1)$

(ii) $\overline{X} \sim N(\mu, \frac{\sigma^2}{n}) \implies \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

(iii) $\frac{\sum_{i=1}^n (X_i-\mu)^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2_{(n)}$ *(from points (i) and 3 above)*

(iv) $\frac{n(\overline{X}-\mu)^2}{\sigma^2} \sim \chi^2_{(1)}$ *(from points (ii) and 2 above)*

Now, we finally show $\frac{\sum_{i=1}^n (X_i-\overline{X})^2}{\sigma^2} \sim \chi^2_{(n-1)}$. Write:

$$\frac{1}{\sigma^2}\sum_{i=1}^n (X_i-\mu)^2 = \frac{1}{\sigma^2}\left[\sum_{i=1}^n \left[(X_i-\overline{X}) + (\overline{X}-\mu)\right]^2\right] \quad (\text{do } \pm\overline{X})$$

$$= \frac{1}{\sigma^2}\left[\sum_{i=1}^n \left[(X_i-\overline{X})^2 + 2(X_i-\overline{X})(\overline{X}-\mu) + (\overline{X}-\mu)^2\right]\right] \quad (\text{expand square})$$

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2 = \frac{1}{\sigma^2}\left[\sum_{i=1}^{n}(X_i-\overline{X})^2 + 2(\overline{X}-\mu)\sum_{i=1}^{n}(X_i-\overline{X}) + n(\overline{X}-\mu)^2\right]$$

Note that the middle term is zero since:

$$\sum_{i=1}^{n}(X_i-\overline{X}) = \sum_{i=1}^{n}X_i - n\overline{X} = 0$$

Therefore, we can write:

$$\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \frac{n(\overline{X}-\mu)^2}{\sigma^2},$$

where using, respectively, facts (iii) and (iv) above we have

$$\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} \sim \chi^2_{(n)} \quad \text{and} \quad \frac{n(\overline{X}-\mu)^2}{\sigma^2} \sim \chi^2_{(1)}$$

which implies:

$$\frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

(Informally, "$\chi^2_{(n)} =? + \chi^2_{(1)} \implies ? = \chi^2_{(n)} - \chi^2_{(1)} = \chi^2_{(n-1)}$." See lecture for further details.)

3.3. Summary

(Thm 3.1.) If we assume $(X_1, \ldots, X_n) \overset{iid}{\sim} f(x)$ with $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$, then *for any* $f(x)$:

a) $E(\overline{X}) = E(X) = \mu$

b) $\text{Var}(\overline{X}) = \frac{\text{Var}(X)}{n} = \frac{\sigma^2}{n}$

c) $E(S^2) = \text{Var}(X) = \sigma^2$

(Thm 3.3.) If $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu, \sigma^2)$, then:

a) $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$

b) $\overline{X}$ and S^2 are independent

c) $(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{(n-1)}$

3.4. Additional Results

Theorem 3.3. part (a) implies that

$$\frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \sim N(0, 1)$$

- This result is useful to test hypothesis about μ (later)

- *Problem:* usually σ is unknown, we only have S (sample standard deviation).

- So now we ask, what is the distribution of

$$\frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim N(0, 1)?$$

Result 1: If $U \sim N(0, 1)$ and $V \sim \chi^2_{(p)}$ and U and V are independent, then:

$$\frac{U}{\sqrt{V/p}} = t\text{-distribution}(p)$$

- *Informally*:

$$\frac{N(0, 1)}{\sqrt{\frac{\chi^2_{(p)}}{p}}} = t_{(p)}$$

- Sketch of proof:

  * Independence of U and V implies $f_{UV}(u,v) = f_U(u) \cdot f_V(v)$
  * Apply the (bivariate) transformation of variables $t = U/\sqrt{V/p}$ and $W = V$ to get the joint $f_{tW}(t,w)$.
  * Get the marginal $f_t(t)$ from the joint pdf $f_{tW}(t,w)$, which results in the pdf of a t-distribution with p degrees of freedom.

- *Example:* Result 1 is used to show (see details from lecture):

$$(X_1, \ldots, X_n) \sim N(\mu, \sigma^2) \implies \frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim t_{(n-1)}$$

Result 2: Let $U \sim \chi^2_{(r_1)}$ and $V \sim \chi^2_{(r_2)}$ with U and V independent. Then:

$$W = \frac{U/r_1}{V/r_2} \sim F\text{-distribution}(r_1, r_2)$$

- *Informally:*

$$\frac{\chi^2_{(r_1)}}{r_1} \Big/ \frac{\chi^2_{(r_2)}}{r_2} \sim F_{(r_1, r_2)}$$

  In words, W follows an F distribution with r_1 and r_2 degrees of freedom.

- See problem set 7 question 5 for an example (analogous to example for Result 1).

Final Notes: *(Thm. 5.3.8 in textbook)*

(i) If $X \sim F(p,q)$, then:

$$\frac{1}{X} \sim F(q,p)$$

(ii) If $X \sim t_{(q)}$, then:

$$X^2 \sim F_{(1,q)}$$

– Example: Since

$$\sqrt{n}\left(\frac{\overline{X}-\mu}{s}\right) \sim t_{(n-1)}$$

$$\Longrightarrow \frac{n(\overline{X}-\mu)^2}{S^2} \sim F_{(1,n-1)}$$